

PCAP Analysis & Threat Intelligence

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Executive Summary

The supplied capture, **ctf.pcapng**, records a TCP SYN port scan from **192.168.29.10** against **192.168.29.155**, followed by an HTTP file transfer in which a ZIP archive (**dog_flag.jpg.zip**) is served from the scanning host to the target. Reassembling the TCP stream and extracting the archive recovers **dog_flag.jpg**, which contains the CTF flag.

Key Finding	Detail
Scan type	TCP SYN (half-open) port scan / service enumeration
Scanner	192.168.29.10 (private IPv4)
Target	192.168.29.155 (private IPv4)
Scan volume	2,890 SYN probes across 1,427 unique destination TCP ports
Open port observed	23/tcp (SYN-ACK received in packet 207)
File transfer	GET /dog_flag.jpg.zip, HTTP 200, 175,828 bytes (192.168.29.10:8000)
Recovered flag	pkhuyar[dogesh_bhai_jindabad]

1. PCAP Analysis

Traffic was filtered for packets with the SYN flag set and the ACK flag clear, which isolates connection-initiation probes. The parsed capture contains **2,890 SYN probes** directed at **1,427 unique destination ports** on a single host. A high probe count spread across a wide range of ports from one source is the characteristic signature of a TCP SYN scan rather than normal application traffic.

1.1 SYN Probe

```
Display filter : tcp.flags.syn == 1 && tcp.flags.ack == 0
Packet        : 114 (ctf.pcapng)
Frame 114     : TCP 192.168.29.10:65077 → 192.168.29.155:1025
Flags        : 0x002 (SYN)
Scan total   : 2,890 SYN probes
Unique ports  : 1,427
```

Figure 1: SYN probe from the scanner to the target (packet 114).

1.2 Target Response

```
Packet        : 207 (ctf.pcapng)
Frame 207     : TCP 192.168.29.155:23 → 192.168.29.10:65117
Flags        : 0x012 (SYN, ACK)
Meaning       : port 23/tcp on the target is open and accepting connections
```

Figure 2: SYN-ACK from the target, showing port 23 (Telnet) is listening.

1.3 Scan Summary

All values below are derived from the supplied capture:

Attacker / scanner	: 192.168.29.10
Victim / target	: 192.168.29.155
SYN scan probes	: 2,890
Unique destination ports	: 1,427
Flag server	: 192.168.29.10:8000
Flag client	: 192.168.29.155:52822

Figure 3: Values derived from ctf.pcapng.

2. Malicious IP Identification

Role	IP Address	Type	Basis for identification
Attacker / scanner	192.168.29.10	Private (RFC 1918)	Source of 2,890 SYN probes across 1,427 ports; also hosts the HTTP server on port 8000
Victim / target	192.168.29.155	Private (RFC 1918)	Destination of all probes; answered with SYN-ACK on port 23; downloads the ZIP file

Both addresses are internal IPv4 addresses on the same 192.168.29.0/24 network, so the activity is lateral (internal) reconnaissance rather than an external attack. The scanner is identified by the direction and volume of the SYN probes. Because private addresses are not publicly routable, they cannot be checked against public reputation services such as AbuseIPDB.

3. Threat Intelligence

The capture also contains traffic to public Google Cloud addresses. One of them, **34.107.243.93**, was checked against AbuseIPDB.

AbuseIPDB Field	Value
IP address	34.107.243.93
Owner / ISP	Google LLC
ASN	AS396982
Total reports	1,506
Abuse confidence score	83%
Recent report categories	Includes port scanning

Assessment. This reputation data must be read in context. The address belongs to shared Google Cloud / CDN infrastructure, where many unrelated tenants use the same IP ranges over time. Reports against it therefore do not, on their own, show that the hosts in this capture are malicious or compromised. The finding is treated as **low-confidence context**; the confirmed malicious behaviour in this capture is the internal SYN scan described above.

Source: AbuseIPDB lookup for 34.107.243.93. Scores and report counts change over time and reflect the values at the time of checking.

4. Flag Discovery

After the scan, the client (192.168.29.155) requests an archive from the HTTP server running on the scanner (192.168.29.10:8000). The server returns a ZIP file, which was reconstructed from the captured TCP stream.

4.1 HTTP Request

```
Packet : 4455 (ctf.pcapng)

GET /dog_flag.jpg.zip HTTP/1.1
Host: 192.168.29.10:8000
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Referer: http://192.168.29.10:8000/
```

Figure 4: HTTP GET request for dog_flag.jpg.zip (application payload of packet 4455).

4.2 HTTP Response and ZIP Payload

```
Packet 4467      : HTTP response header
Packets 4469-4479 : ZIP payload bytes

HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.14.2
Content-type: application/zip
Content-Length: 175828

First ZIP bytes      : PK 03 04 ... dog_flag.jpg
Reconstructed stream length : 175828 bytes
```

Figure 5: HTTP 200 response and ZIP payload. The reconstructed length matches Content-Length.

The SimpleHTTP/0.6 Python banner is consistent with an ad-hoc server started with Python's built-in http.server module. The PK 03 04 magic bytes confirm the payload is a valid ZIP archive.

4.3 Reconstructed ZIP Contents

```
Stream : 192.168.29.10:8000 → 192.168.29.155:52822 (reassembled from the PCAP)
Archive : flag_recovered.zip

  dog_flag.jpg
  __MACOSX/._dog_flag.jpg

Result : flag image successfully recovered from captured traffic.
```

Figure 6: Contents of the archive after parsing. The __MACOSX entry is macOS metadata created when the archive was zipped and is not relevant to the analysis.

4.4 Recovered Flag Image



Figure 7: dog_flag.jpg extracted from the PCAP.

EXTRACTED FLAG

pkhuyar[dogesh_bhai_jindabad]

5. Indicators and ATT&CK Mapping

Indicator	Value	Note
Scanner IP	192.168.29.10	Source of SYN scan; HTTP server on 8000/tcp
Target IP	192.168.29.155	Scan target; open port 23/tcp
Requested URL	/dog_flag.jpg.zip	Served by SimpleHTTP/0.6 Python/3.14.2
Client source port	52822	HTTP download session
Client User-Agent	Firefox 140.0 (Linux x86_64)	Seen in HTTP GET, packet 4455

Technique	ID	Evidence
Network Service Discovery	T1046	2,890 SYN probes over 1,427 ports
Ingress Tool Transfer (analogous: file transfer over HTTP)	T1105	ZIP archive delivered over HTTP from 192.168.29.10:8000

6. Recommendations

- **Detect scanning:** alert when one source sends SYNs to many distinct ports within a short window (for example through an IDS rule or SIEM threshold).

- **Reduce exposure:** disable Telnet (23/tcp), which is unencrypted, and close any service that is not required; use host firewalls to restrict inbound connections.
- **Segment the network:** limit host-to-host traffic inside the LAN so a single machine cannot enumerate every port on its neighbours.
- **Monitor ad-hoc file servers:** flag unexpected HTTP servers on non-standard ports (such as 8000) that serve archives to other internal hosts.

7. Conclusion

ctf.pcapng provides packet-level evidence of a TCP SYN port scan from 192.168.29.10 against 192.168.29.155, covering 1,427 unique ports, with port 23 found open. The capture also contains an HTTP transfer of dog_flag.jpg.zip from 192.168.29.10:8000 to 192.168.29.155:52822. Reconstructing that TCP stream recovers the ZIP archive and dog_flag.jpg, which contains the hidden flag: pkhuyar[dogesh_bhai_jindabad].

Note: The evidence figures are text renderings of packet data and recovered files taken from the supplied ctf.pcapng.